

BARRHEAD, AB, CanadaWMO: **710610**Lat: **54.095N**Lon: **114.448W**Elev: **2126**StdP: **13.60**Time Zone: **-7.00 (NAM)**Period: **00-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-27.2	-19.8	-33.5	0.9	-27.1	-25.6	1.4	-19.3	20.5	36.2	16.8	28.4	2.4	250	0.634

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.4	83.5	64.0	80.2	62.4	77.1	60.8	66.8	79.3	64.6	76.6	62.7	73.9	6.9	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
62.1	90.3	72.7	59.9	83.2	70.4	57.9	77.5	67.8	32.6	79.0	30.8	76.6	29.3	74.1	73.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.5	15.6	13.3	DB	-38.0	89.1	8.5	3.2	-44.1	91.4	-49.0	93.3	-53.8	95.1	-59.9	97.5	(4)
(5)			WB	-37.4	70.1	8.8	2.1	-43.7	71.6	-48.9	72.8	-53.8	74.0	-60.2	75.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	37.3	12.6	13.9	23.6	38.9	51.3	58.4	62.8	59.9	50.9	38.9	24.0	10.5		
	DBStd	21.61	15.91	14.53	14.19	9.35	8.08	5.40	5.09	5.54	7.43	8.48	12.43	13.95		
	HDD50	5854	1159	1010	817	344	82	4	0	3	76	354	780	1225		
	HDD65	10183	1624	1430	1282	783	428	207	104	172	423	810	1230	1690		
	CDD50	1209	0	0	0	11	121	257	397	310	104	9	0	0		
	CDD65	63	0	0	0	0	2	10	35	15	1	0	0	0		
	CDH74	1417	0	0	0	6	118	241	594	340	111	7	0	0		
	CDH80	304	0	0	0	0	21	46	148	65	23	1	0	0		
	WSAvg	5.6	5.0	4.9	5.7	6.9	7.1	6.1	5.6	4.8	5.4	5.8	5.0	4.4		
	PrecAvg	19.4	0.9	0.6	0.6	1.1	1.9	3.4	3.7	2.5	1.5	0.9	0.9	0.7		
(15) Precipitation	PrecMax	26.6	4.2	1.2	1.1	2.6	4.1	7.8	7.9	6.7	4.0	2.4	2.8	1.9		
	PrecMin	13.4	0.0	0.2	0.2	0.2	0.4	1.7	1.9	1.1	0.2	0.1	0.2	0.1		
	PrecStd	3.4	0.9	0.3	0.3	0.6	0.9	1.3	1.5	1.1	0.8	0.6	0.6	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.9	48.4	54.4	73.8	82.9	85.0	88.8	85.9	83.6	73.0	53.9	45.1		
		MCWB	38.0	38.6	42.7	52.4	57.1	63.0	67.6	65.2	62.0	54.7	43.0	38.2		
	2%	DB	42.3	42.7	49.5	66.0	77.1	80.3	84.0	81.3	76.8	62.3	47.2	39.3		
		MCWB	36.1	35.1	39.1	48.7	55.1	60.6	66.0	64.2	59.3	49.5	38.9	33.8		
	5%	DB	38.4	38.5	45.4	60.2	72.8	76.2	80.4	77.7	71.2	57.1	42.7	35.0		
		MCWB	33.3	32.7	37.0	45.1	53.0	58.7	64.5	62.6	57.1	45.6	36.2	31.2		
	10%	DB	34.4	34.5	41.4	55.5	68.7	72.4	76.9	74.4	65.9	52.6	38.8	30.7		
		MCWB	30.5	30.2	34.7	42.9	51.4	57.1	62.2	61.3	54.1	43.5	34.3	27.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.7	39.3	43.2	52.8	59.1	65.7	70.8	68.6	63.5	55.0	43.8	38.5		
		MCDB	45.3	47.5	53.3	72.0	78.2	79.3	83.4	80.8	80.0	71.3	52.4	44.3		
	2%	WB	36.1	35.7	39.9	49.2	56.5	62.6	67.9	66.0	60.4	50.1	39.5	34.3		
		MCDB	41.8	41.9	48.2	64.2	73.0	76.1	80.1	78.0	74.4	59.6	45.7	38.5		
	5%	WB	33.5	33.1	37.4	46.0	54.6	60.4	65.5	63.9	57.6	46.8	36.9	31.3		
		MCDB	37.9	38.0	44.6	59.0	69.8	73.1	77.7	75.2	69.8	55.0	41.6	35.0		
	10%	WB	30.6	30.3	35.0	43.3	52.4	58.6	63.5	62.0	55.0	44.3	34.6	27.7		
		MCDB	34.3	34.7	40.9	53.8	66.3	69.4	74.5	72.4	63.9	51.3	38.6	30.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	19.8	22.8	22.5	22.4	24.9	22.8	23.4	24.1	24.3	20.8	17.3	19.7		
		MCDBR	21.9	26.5	25.7	32.6	33.4	30.6	30.9	31.4	34.9	30.1	21.5	22.0		
		MCWBR	16.5	19.9	16.7	17.3	15.1	13.8	14.7	16.0	18.3	17.4	14.5	17.7		
	5% WB	MCDBR	21.6	25.6	24.6	31.4	30.0	26.5	28.3	28.6	32.7	28.1	20.0	21.3		
		MCWBR	16.8	19.5	16.3	17.2	13.9	12.7	14.4	15.5	17.5	17.0	14.3	17.4		
(40) Clear-Sky Solar Irradiance	taub		0.249	0.269	0.300	0.330	0.347	0.357	0.359	0.361	0.314	0.283	0.254	0.239		
	taud		2.324	2.337	2.347	2.380	2.379	2.362	2.411	2.355	2.494	2.555	2.454	2.292		
	Ebn at Noon		233	266	281	285	285	282	279	272	274	262	234	212		
	Edh at Noon		17	24	30	33	35	36	34	34	26	19	15	14		
(44) All-Sky Solar Radiation	RadAvg		281	601	1059	1462	1760	1856	1876	1559	1080	611	294	202		
	RadStd		23	45	80	128	137	132	118	113	112	67	28	18		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (49 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page